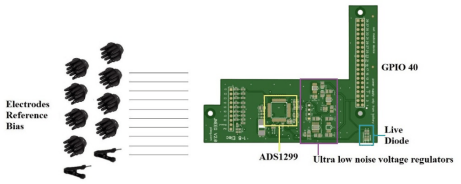
CONNECTOR PINOUT & ELECTRODE GUIDE · SKU CQA240229JN

Electrodes, Header & Jetson Pins

How to connect 8 electrodes, the reference and the bias to JNEEG, the default 10-20 montage, the five Jetson Nano header pins the shield uses, and the LEDs and test points on the board.



8	REF	BIAS	5	2.54
Electrode channels	Ear clip · common 0 µV	Ear clip · noise rejection	Header signal pins	mm header pitch

01 ELECTRODE INPUTS

Input	Count	Electrode	Purpose
Channels 1–8	8	Dry or wet Ag/AgCl, in the cap	Voltage between each electrode and the reference
Reference	1	Ear clip on one earlobe	Common 0 µV point for all 8 channels
Bias	1	Ear clip on the other earlobe	Drives the body to cancel common-mode noise such as 50/60 Hz hum

02 ELECTRODE HEADER J3

Square pad = pin 1 (E1). 2.54 mm pitch, single row.

E1	E2	E3	E4	E5	E6	E7	E8	Biasout	Ref
pin 1	pin 2	pin 3	pin 4	pin 5	pin 6	pin 7	pin 8	pin 9	pin 10
Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7	Channel 8	Bias ear clip	Reference ear clip

Figure 1 — Electrode header as printed on the board (“1-8 Elec”), top view, E1 on the left.

> **NOTE Reference and bias clips matter most.**
> A loose ear clip makes every channel noisy at once. Clean the earlobes and make sure both clips sit firmly. Keep electrode leads away from the Jetson fan and heatsink.

03 LEDS AND TEST POINTS

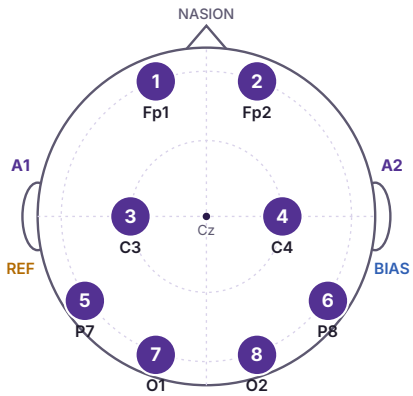
LED	Meaning
D1 “power”	Shield powered
D2 “connection”	ADS1299 connected / running

J1 pin	Expected	Use
−2.5V	≈ −2.5 V	Supply check
2.5V	≈ +2.5 V	Supply check
3.2V	≈ 3.2 V	Supply check
gnd	0 V	Meter reference

Measure J1 with a multimeter only, with no electrodes on anyone. Do not draw current from the test points.

For research and education only. Not a medical device. Specifications subject to change.

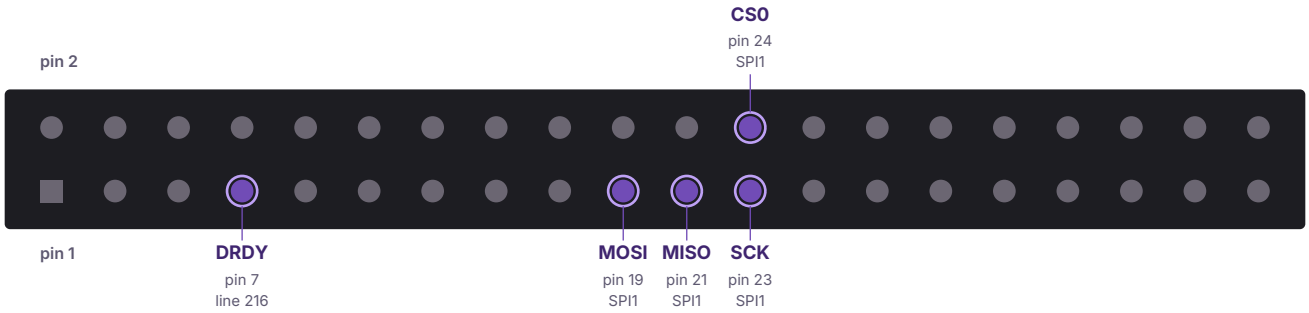
04 DEFAULT MONTAGE (10-20 SYSTEM)



Ch	Position	Area	Good for
1	Fp1	Front left	Blinks, eye movement
2	Fp2	Front right	Blinks, eye movement
3	C3	Centre left	Motor activity (right hand)
4	C4	Centre right	Motor activity (left hand)
5	P7	Back left, side	Parietal / temporal
6	P8	Back right, side	Parietal / temporal
7	O1	Back left	Alpha rhythm, vision
8	O2	Back right	Alpha rhythm, vision

The same layout as the PiEEG-8 default, so recordings can be compared. Any other positions work — note which channel went where.

05 JETSON NANO 40-PIN HEADER



Odd pins: bottom row · even pins: top row · power and ground are also taken from the header

Header pin	Signal	Linux	Direction
19	SPI MOSI	spidev0.0	Jetson → ADS1299
21	SPI MISO (DOUT)	spidev0.0	ADS1299 → Jetson
23	SPI SCK	spidev0.0	Jetson → ADS1299
24	SPI CS0	spidev0.0	Jetson → ADS1299
7	DRDY (active low)	gpiochip0 line 216 · Jetson.GPIO board pin 7	ADS1299 → Jetson

SPI is the `spi1` function in `jetson-io` and appears as `/dev/spidev0.0`. The shield also takes 5 V, 3.3 V and GND from the header; other pins stay free.

06 RELATED DOCUMENTS

JN-DS-001
Product Datasheet

JN-MD-002
Mechanical & Dimensional Drawing

JN-PCB-003
PCB & Manufacturing Specification

JN-QSG-005
Quick Start & Safety Guide

JN-SW-006
Data Interface & Software Guide

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